A hybrid neural simulation-based inference method for LHC analyses

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Neural simulation-based inference based on classifier estimates of density ratios has made unbinned, high-dimensional likelihood-ratio inference practical for analyses with many physics processes and nuisance parameters. Its ratio-only formulation, however, does not provide an explicit normalized event density or a direct generative model. Consequently, expected-data calculations and frequentist pseudo-experiments must be represented by very large weighted simulated samples, which can dominate the computational cost of profiling, uncertainty decomposition, and Neyman constructions. Normalizing flows provide normalized densities and efficient sampling, but small absolute-density errors can be consequential when a small signal is inferred in a much larger background.

We propose *hybrid neural simulation-based inference*. A normalizing flow is trained on a balanced, physically representative reference distribution. Events sampled from that learned reference—rather than the finite simulation used to train the flow—form the denominator sample for classifier-based estimates of each process-to-reference density ratio. A process density is then represented as the product of the tractable reference-flow density and its learned ratio. This construction preserves the discriminative precision of density-ratio estimation while making every component of the event intensity tractable and sampleable. The same fixed reference sample supplies normalized importance weights for process generation and a deterministic weighted, unbinned Asimov dataset. We derive the extended likelihood, the sampling prescription, and the Asimov construction, and demonstrate the method in a five-dimensional Gaussian-mixture measurement with a small signal in a large background. This proof of concept establishes a route from parametrized density-ratio inference to a generative, likelihood-based analysis without requiring an independently precise absolute-density estimator for every process.

I. INTRODUCTION

Likelihood-based measurements at the Large Hadron Collider (LHC) conventionally compress reconstructed events into one or a few binned observables. This construction is robust and computationally convenient, but the compression and binning can discard information when the likelihood depends nonlinearly on parameters or when no low-dimensional summary is sufficient over the full parameter space. Neural simulation-based inference (NSBI) instead uses high-dimensional simulated events to estimate likelihoods or likelihood ratios directly in reconstructed feature space [1–3].

The ATLAS Collaboration developed a practical classifier-based NSBI formalism for LHC parameter estimation [4]. It was deployed in the measurement of off-shell Higgs-boson production in the $H^* \to ZZ \to 4\ell$ final state [5]. The formalism uses neural estimates of density ratios and an analytic factorization of the dependence on parameters of interest (POIs) and nuisance parameters. It can therefore describe dozens of POIs and hundreds of nuisance parameters without densely sampling that complete parameter space during neural-network training. We refer to this construction as *parametrized NSBI*; it has also been described as a factorized or semi-parametric approach [6].

Density ratios are sufficient for inference based on a profile likelihood ratio. They are also often easier to estimate to the required precision than absolute densities. A reference sample can be designed to resemble the relevant physics, sharing resonant structure, tails, and detector effects with the target samples. The resulting ratios vary more mildly than either absolute density, and they can be learned discriminatively with a binary cross-entropy objective. This numerical advantage was central to the precision achieved in the ATLAS studies.

A ratio-only representation nevertheless has an important operational limitation: it is not itself a generative density model. An unbinned Asimov dataset must be approximated by a large weighted Monte Carlo (MC) sample, and pseudo-experiments must be constructed by resampling such a sample. The former makes repeated profiling and impact calculations costly; the latter requires an MC pool much larger than a typical pseudo-experiment [4]. These difficulties do not invalidate ratio-based inference, but they make some standard LHC statistical workflows expensive.

Normalizing flows address the complementary problem. They provide both a normalized, tractable density and efficient event generation [7–10]. The difficulty is precision: two separately trained process flows can each pass many marginal checks while their residual, unrelated absolute-density errors bias the process-density ratio that controls a small-signal measurement. Recent work has developed factorized parameter-dependent flows [11], but the required absolute-density accuracy remains a central question when the signal is orders of magnitude smaller than the background.

This paper combines the two representations. A flow is used to build a normalized, sampleable reference density, while classifiers learn the precision-critical ratios of each process to that exact learned reference. The reference flow is a proposal distribution, not an uncorrected model of every process. If the ratio estimators are exact, local reference-flow error cancels algebraically. What the flow must provide is common support and efficient coverage. The resulting *hybrid NSBI* construction supplies a tractable likelihood, direct weighted or resampled event generation, and a deterministic unbinned Asimov measure.

The remainder of this paper is organized as follows. Section II reviews parametrized NSBI and its practical limitations. Section III introduces the hybrid construction. Section IV develops weighted sampling and pseudo-experiments, and Section V derives the weighted unbinned Asimov dataset. Section VI describes a five-dimensional proof-of-concept experiment corresponding to the conference exercise. We discuss validation requirements and extensions in Section VII and conclude in Section VIII.

II. PARAMETRIZED NEURAL SIMULATION-BASED INFERENCE

A. Extended unbinned likelihood

Let $\boldsymbol{x} \in \mathbb{R}^d$ denote the reconstructed observables for an event and let j index physics components. For parameters $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\alpha})$, including POIs $\boldsymbol{\mu}$ and nuisance parameters $\boldsymbol{\alpha}$, define the differential event intensity

$$\nu(\boldsymbol{x} \mid \boldsymbol{\theta}) = \sum_j \lambda_j(\boldsymbol{\theta}) p_j(\boldsymbol{x} \mid \boldsymbol{\theta}), \tag{1}$$

where p_j is normalized to unity and λ_j is the expected yield. Its integral is

$$\Lambda(\boldsymbol{\theta}) = \int \nu(\boldsymbol{x} \mid \boldsymbol{\theta})\, \mathrm{d}\boldsymbol{x} = \sum_j \lambda_j(\boldsymbol{\theta}). \tag{2}$$

Up to factors independent of $\boldsymbol{\theta}$, the extended unbinned likelihood for n observed events is

$$\mathcal{L}(\boldsymbol{\theta}) = e^{-\Lambda(\boldsymbol{\theta})} \prod_{i=1}^{n} \nu(\boldsymbol{x}_i \mid \boldsymbol{\theta}) \prod_k \pi_k(a_k \mid \alpha_k), \tag{3}$$

where π_k are constraint terms for auxiliary measurements a_k. The profile-likelihood-ratio statistic for a POI value $\boldsymbol{\mu}$ is

$$t_{\boldsymbol{\mu}} = -2 \log \frac{\mathcal{L}(\boldsymbol{\mu}, \widehat{\widehat{\boldsymbol{\alpha}}}_{\boldsymbol{\mu}})}{\mathcal{L}(\widehat{\mu}, \widehat{\boldsymbol{\alpha}})}. \tag{4}$$

B. Classifiers as density-ratio estimators

Consider a classifier trained with equal total class weight to distinguish events from densities p_1 and p_0. At the population minimum of the binary cross-entropy loss, its score is

$$s^*(\boldsymbol{x}) = \frac{p_1(\boldsymbol{x})}{p_1(\boldsymbol{x}) + p_0(\boldsymbol{x})}. \tag{5}$$

The density ratio follows from the odds [1],

$$r(\boldsymbol{x}) = \frac{p_1(\boldsymbol{x})}{p_0(\boldsymbol{x})} = \frac{s^*(\boldsymbol{x})}{1 - s^*(\boldsymbol{x})}. \tag{6}$$

Balanced class weights are important: otherwise the class-prior odds must be included explicitly. In practice, ensembles, calibration tests, and reweighting closure tests are used to control finite-sample, optimization, and model-capacity effects [4].

Choose a fixed reference density $p_{\rm ref}(\boldsymbol{x})$ whose support covers all relevant components, and define

$$r_j(\boldsymbol{x} \mid \boldsymbol{\theta}) = \frac{p_j(\boldsymbol{x} \mid \boldsymbol{\theta})}{p_{\rm ref}(\boldsymbol{x})}. \tag{7}$$

The intensity may then be written

$$\nu(\boldsymbol{x} \mid \boldsymbol{\theta}) = p_{\rm ref}(\boldsymbol{x}) \sum_j \lambda_j(\boldsymbol{\theta}) r_j(\boldsymbol{x} \mid \boldsymbol{\theta}). \tag{8}$$

Because the reference is parameter independent, $\sum_i \log p_{\rm ref}(\boldsymbol{x}_i)$ cancels from every likelihood ratio. Hence the reference need not have a tractable density for the inference in Eq. (4); only the component-to-reference ratios are required.

C. Factorized parameter dependence

Directly training a single conditional estimator throughout a space with many POIs and nuisance parameters generally requires dense simulated coverage of that space. Parametrized NSBI instead exploits the analytic structure of an LHC likelihood. A useful schematic form is

$$\begin{aligned} \frac{p(\boldsymbol{x} \mid \boldsymbol{\mu}, \boldsymbol{\alpha})}{p_{\rm ref}(\boldsymbol{x})} &= \frac{1}{\Lambda(\boldsymbol{\mu}, \boldsymbol{\alpha})} \sum_j f_j(\boldsymbol{\mu}) \lambda_j r_j^0(\boldsymbol{x}) \\ &\quad \times \prod_k G_{jk}(\alpha_k) g_{jk}(\boldsymbol{x}, \alpha_k). \end{aligned} \tag{9}$$

Here f_j describes known POI dependence, G_{jk} is a yield interpolation, $r_j^0 = p_j^0/p_{\rm ref}$ is a nominal process-to-reference ratio, and $g_{jk} = p_j(\boldsymbol{x} \mid \alpha_k)/p_j^0(\boldsymbol{x})$ describes a systematic shape variation. The individual ratios can be trained on nominal and varied samples; their parameter dependence is supplied by low-dimensional analytic interpolation. This is a simplification of fully conditional, simulator-assisted methods [2, 3], but it matches how systematic variations are made available in full LHC analyses and scales to large nuisance-parameter models [4, 6].

D. Operational limitations of a ratio-only model

The cancellation of $p_{\rm ref}$ in Eq. (8) is an advantage for parameter estimation, but it also means that the construction does not specify a normalized density that can be evaluated or sampled. Two consequences are particularly relevant.

First, an Asimov dataset is meant to return the generating parameter without statistical fluctuation and to encode median expected sensitivity [12]. In a binned likelihood, the expected count is placed in every bin. In an unbinned ratio-only analysis, one instead uses a large weighted MC sample as numerical quadrature for the expected event measure. The approximation improves as this sample grows, but evaluating hundreds of neural networks on millions of events and repeating the calculation for profiles or impact decompositions can be expensive. The ATLAS implementation reports likelihood maximizations ranging from minutes and gigabytes to hours and hundreds of gigabytes as the selected sample grows from thousands to millions of events [4].

Second, a Neyman construction requires samples from the model at each tested parameter point. A ratio alone does not provide a direct sampler. A valid workaround is to draw with replacement from a much larger positive-weight reference pool, with probabilities proportional to event weights or learned ratios. The ATLAS procedure assigns each reference event a Poisson-distributed integer multiplicity with mean equal to its Asimov weight [4]. This is statistically sound for the discrete MC approximation but requires a pool much larger than an individual pseudo-experiment, and the pool remains the resolution limit of the generator.

These are computational and representational limitations, not failures of density-ratio inference. They motivate adding a generative reference model without surrendering the ratio precision that made parametrized NSBI practical.

III. HYBRID NEURAL SIMULATION-BASED INFERENCE

A. A learned reference as a proposal density

Let $q_\phi(\boldsymbol{x})$ be a normalized normalizing-flow density trained on a reference sample. For a signal-plus-background problem, a useful training target is a balanced mixture after the analysis selection,

$$p_{\rm mix}(\boldsymbol{x}) = \tfrac{1}{2} p_S(\boldsymbol{x}) + \tfrac{1}{2} p_B(\boldsymbol{x}). \tag{10}$$

The equal mixture is deliberate. A reference with physical normalization would be nearly identical to the background when $S/B \ll 1$ and would spend little capacity in signal-rich regions. The balanced target instead asks the flow to cover both process phase spaces comparably.

After training, a large, dedicated sample is drawn from q_ϕ. Crucially, this *flow-generated* sample, not the original MC mixture in Eq. (10), is used as the denominator class for the density-ratio classifiers. For every physics component j, the classifier therefore estimates

$$r_j^\phi(\boldsymbol{x}) = \frac{p_j(\boldsymbol{x})}{q_\phi(\boldsymbol{x})}. \tag{11}$$

The hybrid process density is

$$\widehat{p}_j(\boldsymbol{x}) = q_\phi(\boldsymbol{x})\widehat{r}_j^\phi(\boldsymbol{x}). \tag{12}$$

This ordering is the key observation. Suppose the flow approximates the intended mixture imperfectly, $q_\phi \neq p_{\mathrm{mix}}$. An exact classifier trained against samples from q_ϕ still learns p_j/q_ϕ, and Eq. (12) returns p_j exactly. Local flow error is corrected by the ratio. The flow is not required to be an independently precise model of p_S or p_B; it must be evaluable, sampleable, and nonzero wherever any target density is nonzero.

The support condition is essential. If $p_j(\boldsymbol{x}) > 0$ where $q_\phi(\boldsymbol{x})$ is negligibly small, the ratio becomes large, classifier learning is difficult, and importance sampling has low effective sample size. A balanced, physics-informed mixture, broad flow tails, and explicit support diagnostics are therefore part of the statistical design rather than optional training refinements.

B. Tractable event intensity and likelihood

With hybrid component densities, the event intensity becomes

$$\widehat{\nu}(\boldsymbol{x} \mid \boldsymbol{\theta}) = q_\phi(\boldsymbol{x}) \sum_j \lambda_j(\boldsymbol{\theta})\widehat{r}_j^\phi(\boldsymbol{x} \mid \boldsymbol{\theta}). \tag{13}$$

Every term is now tractable. Define $h_{\boldsymbol{\theta}}(\boldsymbol{x}) = \sum_j \lambda_j(\boldsymbol{\theta})\widehat{r}_j^\phi(\boldsymbol{x} \mid \boldsymbol{\theta})$. For a fixed reference flow, the contribution $\sum_i \log q_\phi(\boldsymbol{x}_i)$ remains independent of $\boldsymbol{\theta}$ and cancels from profile likelihood ratios. Thus inference retains the original ratio-based numerical structure,

$$\begin{aligned} -2\log\mathcal{L}(\boldsymbol{\theta}) = 2\Lambda(\boldsymbol{\theta}) &- 2\sum_i \log q_\phi(\boldsymbol{x}_i) \\ &- 2\sum_i \log h_{\boldsymbol{\theta}}(\boldsymbol{x}_i) - 2\sum_k \log \pi_k, \end{aligned} \tag{14}$$

while the explicit q_ϕ factor is available whenever an absolute density is needed.

The same factorized POI and nuisance-parameter model in Eq. (9) can be used after replacing the implicit reference by q_ϕ. Consequently, the hybrid proposal is orthogonal to whether the parameter dependence is described by analytic interpolation, a conditional neural network, a Gaussian process, or another nonparametric estimator. Developments aimed at relaxing the factorization assumptions can be incorporated without changing the reference-density layer [11, 13].

C. Finite-sample normalization

For an exact normalized ratio,

$$\mathbb{E}_{\boldsymbol{x}\sim q_\phi}[r_j^\phi(\boldsymbol{x})] = 1. \tag{15}$$

Finite classifiers generally violate this equality slightly. Given a fixed reference sample $\{\boldsymbol{z}_m\}_{m=1}^M \sim q_\phi$, we impose

$$\widetilde{r}_j(\boldsymbol{z}_m) = \frac{\widehat{r}_j^\phi(\boldsymbol{z}_m)}{M^{-1}\sum_{\ell=1}^M \widehat{r}_j^\phi(\boldsymbol{z}_\ell)}. \tag{16}$$

This guarantees exact component normalization on the empirical reference quadrature. It does not repair local ratio bias, so calibration and multidimensional reweighting tests remain necessary. It does, however, decouple local-shape validation from a trivial normalization offset and is what makes the discrete Asimov stationarity exact in Section V.

IV. SAMPLING AND NEYMAN CONSTRUCTION

The fixed flow reference sample can represent any component by importance weights

$$w_m^{(j)} = \frac{\widetilde{r}_j(\boldsymbol{z}_m)}{M}, \qquad \sum_{m=1}^{M} w_m^{(j)} = 1. \tag{17}$$

Weighted histograms or expectations calculated with these weights approximate integrals over p_j. To generate an unweighted component event, one may sample an index m with probability $w_m^{(j)}$. Alternatively, new proposal points can be drawn from the flow and resampled using freshly evaluated ratios. Rejection sampling is also possible when a reliable finite upper bound on the ratio is available.

For a pseudo-experiment at parameter point $\boldsymbol{\theta}$, one first draws the component counts independently,

$$n_j \sim \mathrm{Pois}(\lambda_j(\boldsymbol{\theta}))\,, \tag{18}$$

and then draws n_j indices according to $w_m^{(j)}$. Concatenating the selected reference events gives an unweighted pseudo-dataset from the empirical hybrid model. Repeating the likelihood fit yields the sampling distribution of $t_{\boldsymbol{\mu}}$ needed for a Neyman construction [14].

The quality of this generator can be summarized by the effective sample size

$$M_{\mathrm{eff}}^{(j)} = \frac{(\sum_m \widetilde{r}_j)^2}{\sum_m \widetilde{r}_j^2}. \tag{19}$$

Large ratio tails reduce M_{eff} even when M itself is large. Monitoring the effective sample size, upper ratio quantiles, and duplicate rates in toys provides a direct diagnostic of reference coverage. Unlike a fixed original MC pool, however, a flow proposal can generate additional independent reference points whenever the quadrature or toy pool must be enlarged.

V. A WEIGHTED UNBINNED ASIMOV DATASET

A. Expected event measure

An Asimov dataset is most naturally an expected event *measure*, not a particular random list of unweighted events. At a generating point $\boldsymbol{\theta}_A$, define

$$\mathrm{d}N_A(\boldsymbol{x}) = \widehat{\nu}(\boldsymbol{x} \mid \boldsymbol{\theta}_A)\,\mathrm{d}\boldsymbol{x}. \tag{20}$$

For any integrable function f,

$$\begin{aligned}\int f(\boldsymbol{x})\,\mathrm{d}N_A(\boldsymbol{x}) &= \mathbb{E}_{q_\phi}\left[f(\boldsymbol{x})\sum_j \lambda_j(\boldsymbol{\theta}_A)\widetilde{r}_j(\boldsymbol{x} \mid \boldsymbol{\theta}_A)\right] \\ &\simeq \sum_{m=1}^{M} w_m^A f(\boldsymbol{z}_m),\end{aligned} \tag{21}$$

with deterministic event weights

$$w_m^A = \frac{1}{M}\sum_j \lambda_j(\boldsymbol{\theta}_A)\widetilde{r}_j(\boldsymbol{z}_m \mid \boldsymbol{\theta}_A). \tag{22}$$

The extended part is included automatically:

$$\sum_{m=1}^{M} w_m^A = \sum_j \lambda_j(\boldsymbol{\theta}_A) = \Lambda(\boldsymbol{\theta}_A), \tag{23}$$

where the first equality is exact on the empirical quadrature when every component ratio is normalized as in Eq. (16).

The weighted Asimov log likelihood is

$$\log \mathcal{L}_A(\boldsymbol{\theta}) = -\Lambda(\boldsymbol{\theta}) + \sum_{m=1}^{M} w_m^A \log \widehat{\nu}(\boldsymbol{z}_m \mid \boldsymbol{\theta}) + \log \pi(\boldsymbol{a}_A \mid \boldsymbol{\alpha}), \tag{24}$$

with auxiliary data set to their expectations. In the continuous limit, the expected score vanishes at $\boldsymbol{\theta}_A$ under standard regularity conditions. On the finite empirical quadrature, the same property holds exactly for yield-linear components normalized on that quadrature.

B. One signal-strength parameter

For the demonstration, consider

$$\widehat{\nu}(\boldsymbol{x} \mid \mu) = q_\phi(\boldsymbol{x}) \left[\mu \lambda_S \widetilde{r}_S(\boldsymbol{x}) + \lambda_B \widetilde{r}_B(\boldsymbol{x})\right]. \tag{25}$$

At μ_A, define

$$Q_m = \frac{\lambda_S \widetilde{r}_S(\boldsymbol{z}_m)}{\lambda_B \widetilde{r}_B(\boldsymbol{z}_m)}. \tag{26}$$

Dropping terms independent of μ, the Asimov negative log likelihood is

$$\widetilde{\mathrm{NLL}}_A(\mu) = 2\left[\mu\lambda_S - \sum_m w_m^A \log(1 + \mu Q_m)\right]. \tag{27}$$

Because

$$w_m^A \frac{Q_m}{1 + \mu_A Q_m} = \frac{\lambda_S \widetilde{r}_S(\boldsymbol{z}_m)}{M}, \tag{28}$$

the score at the generating point is

$$\left.\frac{\partial \widetilde{\mathrm{NLL}}_A}{\partial \mu}\right|_{\mu_A} = 2\lambda_S \left[1 - \frac{1}{M}\sum_m \widetilde{r}_S(\boldsymbol{z}_m)\right] = 0. \tag{29}$$

The second derivative is nonnegative. Thus the empirical weighted Asimov dataset has its global minimum at μ_A in the physical region, up to numerical optimizer tolerance.

C. Expected uncertainty and toy distributions

Following Cowan *et al.* [12], evaluate the discovery statistic q_0 on an Asimov dataset generated at $\mu_A > 0$:

$$q_{0,A} = -2 \log \frac{\mathcal{L}_A(0)}{\mathcal{L}_A(\widehat{\mu}_A)}. \tag{30}$$

In the one-parameter Wald approximation,

$$q_{0,A} \simeq \frac{\mu_A^2}{\sigma_A^2}, \qquad \sigma_A = \frac{\mu_A}{\sqrt{q_{0,A}}}, \qquad Z_A = \sqrt{q_{0,A}}. \tag{31}$$

The pseudo-experiment distribution of $\widehat{\mu}$ can be compared with the Gaussian prediction of width σ_A. For an unconstrained two-sided statistic, t_0 approaches a noncentral χ^2 distribution with one degree of freedom and noncentrality $q_{0,A}$. With the physical constraint $\widehat{\mu} \geq 0$, the discovery statistic is instead represented asymptotically by

$$q_0 = \left[\max(0, Y)\right]^2, \qquad Y \sim \mathcal{N}(\sqrt{q_{0,A}}, 1), \tag{32}$$

which includes a point mass $\Phi(-\sqrt{q_{0,A}})$ at zero.

For multiple normalization parameters or nuisance parameters, w_m^A is still given by Eq. (22). The symbol differentiated in the score is simply a component of $\boldsymbol{\theta}$. The expected score vector vanishes at $\boldsymbol{\theta}_A$, and the inverse expected Hessian gives the asymptotic covariance matrix after profiling. The construction is therefore not restricted to one parameter; the one-dimensional derivation merely makes the cancellation transparent.

RESULT PLACEHOLDER

Reference-flow closure after preselection.
Recommended content: one-dimensional projections and joint/correlation closure,
accompanied by the learned-log-density CDF comparison.

FIG. 1. Closure of the balanced reference flow against an independent post-selection MC sample. The final figure should demonstrate both marginal and joint-density agreement; matching only one-dimensional projections is insufficient.

VI. FIVE-DIMENSIONAL DEMONSTRATION

We implement the hybrid construction in Exercise 5 of the ML4HEP 2026 conference tutorial [6, 15]. The present section records the frozen analysis design. Numerical values and final figures will be inserted after the long training run is complete; placeholders are deliberately retained so that the text does not imply results that have not yet been reviewed.

A. Samples and preselection

The toy measurement contains signal and background events described by five latent variables. Each process is a correlated Gaussian mixture with two process-specific modes and a common broad component. Independent Gaussian detector smearing produces the reconstructed inputs $\boldsymbol{x} = (x_1, \ldots, x_5)$. The common broad component guarantees overlapping analytic support and makes exact reconstructed densities available for validation. The generated samples contain 10^8 background and 2×10^7 signal events, with inclusive expected yields

$$\lambda_B^{\mathrm{incl}} = 10^6, \qquad \lambda_S^{\mathrm{incl}} = 1.1 \times 10^3. \tag{33}$$

A classifier preselection is trained with at most 5×10^5 events per class using three hidden layers of 256 nodes. The threshold on its estimate of p_S/p_B is selected so that the expected post-selection ratio is approximately $B/S = 250$. The preselection is not part of the final likelihood; it defines the phase-space region in which all densities are normalized. Deterministic row hashes produce disjoint preselection-training, model-training, and final-evaluation partitions, and the parquet samples are streamed so the full event sets need not reside in memory.

B. Reference flow

After preselection, 5×10^5 signal and 5×10^5 background events are concatenated with equal component normalization. A normalizing flow is trained by unweighted maximum likelihood on this balanced sample. The chosen model has ten quadratic-spline coupling transforms. Each conditioner has four hidden layers with 1024 features; the splines use 16 bins over the standardized interval $[-5, 5]$ and linear tails outside it. Training uses batches of 2048, an initial learning rate of 10^{-4}, and at most 70 epochs with early stopping.

Because a smooth flow can place small probability outside the hard preselection boundary, generated events are passed through the same preselection classifier and rejected if they fail. Five million accepted events are retained as a fixed unweighted reference sample. The corresponding accepted reference density includes the flow acceptance normalization. Validation includes one- and two-dimensional closure, correlation closure, the distribution and cumulative distribution of the learned log density, and a direct comparison with the analytic balanced-mixture density.

RESULT PLACEHOLDER

Signal/reference and background/reference calibration and reweighting closure, followed by hybrid-density versus analytic-truth comparisons. Emphasize whether multiplication by the learned ratio removes local reference-flow bias.

FIG. 2. Validation of the classifier ratios and reconstructed hybrid densities. The process densities are evaluated as $\widehat{p}_j = q_\phi \widetilde{r}_j$. The numerical comparison will be filled from the completed Exercise 5 run.

C. Process-to-reference ratios

Two classifier ensembles estimate p_S/q_ϕ and p_B/q_ϕ. Each class contains five million events: selected target MC in the numerator and flow-generated reference events in the denominator. Each ensemble has four independently initialized dense networks, each with four hidden layers of 1024 nodes, trained for at most 50 epochs with batches of 4096 and an initial learning rate of 10^{-3}. Class weights are normalized separately, and no post-training histogram calibration is applied. The arithmetic mean of the four density-ratio predictions is used in the likelihood. The final ratios are normalized on the fixed five-million-event reference sample according to Eq. (16).

Validation is performed on a common held-out partition. Score calibration tests the classifier interpretation in Eq. (6); process projections reweighted from the reference test $q_\phi \widehat{r}_j$; and the reconstructed log densities are compared with analytic truth. The last comparison is decisive: a residual error in the reference flow should be corrected by the classifier trained against that exact flow distribution.

D. Inference, Asimov dataset, and toys

The post-selection measurement has one POI, the signal strength μ, with intensity given by Eq. (25). A JAX-differentiable extended negative log likelihood is minimized with Minuit under $\mu \geq 0$. The hybrid profile-likelihood scan is compared with the scan obtained from the analytic post-selection densities on the same weighted evaluation sample. This controls finite evaluation-sample fluctuations and isolates learned-model bias.

Before pseudo-experiments are generated, the fixed reference sample is converted into the weighted Asimov dataset at $\mu_A = 1$ using Eq. (22). Its total weight is exactly $\lambda_S + \lambda_B$ on the empirical quadrature, and Eq. (29) predicts a fitted minimum at one. The value $q_{0,A}$ gives the expected σ_A and median significance through Eq. (31).

Finally, ensembles of pseudo-experiments are generated at $\mu = 0$ and $\mu = 1$. Independent Poisson signal and background counts are drawn, and reference indices are sampled with probabilities proportional to the normalized signal and background ratio weights. The demonstration currently uses 100 toys per hypothesis. It compares the distribution of $\widehat{\mu}$ with the Asimov/Wald prediction, the central $t_{\mu_{\mathrm{true}}}$ distribution with its asymptotic form, and the q_0 distribution at $\mu = 1$ with both the boundary-aware prediction in Eq. (32) and the ordinary noncentral χ^2 approximation.

VII. DISCUSSION AND OUTLOOK

Hybrid NSBI changes the role of absolute density estimation. Training an independent flow for every process asks each model to be accurate enough that the ratio of two imperfect densities supports a precision measurement. The hybrid approach asks one flow to cover a reference phase space and then learns every precision-critical correction discriminatively. Because the denominator examples are drawn from the trained flow, classifier ratios can correct the flow's local density distortions. At the same time, the explicit q_ϕ factor turns the original ratio representation into a normalized density model.

RESULT PLACEHOLDER

Left: hybrid and analytic profile-likelihood scans, including fitted minima.
Center: weighted-Asimov fit at $\mu_A = 1$ and extraction of $q_{0,A}$ or σ_A.
Right: toy $\widehat{\mu}$ and q_0 distributions compared with the Asimov/Wald predictions.

FIG. 3. Inference-level validation of hybrid NSBI. At minimum, the final result should show an unbiased hybrid fit relative to analytic truth, exact closure of the empirical weighted Asimov fit at $\mu_A = 1$, and agreement between the Asimov prediction and pseudo-experiment sampling distributions within their expected finite-toy precision.

TABLE I. Results to be filled after the completed Exercise 5 run is reviewed. The analytic comparison must use the same post-selection yields and evaluation measure as the hybrid fit.

Quantity	Value
Post-selection λ_S	*pending*
Post-selection λ_B	*pending*
Raw $\mathbb{E}_{q_\phi}[\widehat{r}_S]$	*pending*
Raw $\mathbb{E}_{q_\phi}[\widehat{r}_B]$	*pending*
Hybrid $\widehat{\mu}$ on validation measure	*pending*
Analytic $\widehat{\mu}$ on validation measure	*pending*
Weighted-Asimov $\widehat{\mu}_A$	*pending*
$q_{0,A}$	*pending*
$\sigma_A = 1/\sqrt{q_{0,A}}$	*pending*
Toy width or RMS at $\mu = 1$	*pending*

This construction addresses the two operational limitations discussed in Section II D. The flow can generate arbitrarily many independent reference points, so numerical quadrature for profiles, impacts, and Asimov calculations is not capped by the size of the original reference MC. The normalized ratio weights turn the same proposal into process-specific samplers, enabling pseudo-experiments and Neyman constructions without a new generative model for every parameter point. The original parametrized nuisance model can be retained, while more flexible nonparametric parameter dependence can be added at the ratio layer.

Several limitations remain and should be treated as part of the method definition.

- *Support and importance-weight variance.* A flow that misses a relevant region creates extreme ratios and low effective sample size. Balanced-mixture design, tail modeling, ratio-tail monitoring, and process-wise effective sample sizes are mandatory diagnostics.

- *Classifier misspecification.* Empirical normalization fixes only the integral. Calibration, reweighting, ensemble stability, and inference-level closure against independent samples remain necessary. Neyman construction is especially valuable when residual neural bias could invalidate simple asymptotic assumptions.

- *Finite reference quadrature.* The weighted Asimov dataset is exactly stationary for the empirical hybrid model, not automatically for the unknown data-generating process. Reference-sample enlargement and repeated-flow or repeated-ratio studies can separate quadrature, flow, and classifier uncertainties.

- *Negative-weight and interference components.* Individual signed contributions are not probability densities and cannot be sampled directly. A practical implementation must use positive physical combinations as sampling channels, while retaining analytic signed coefficients in the intensity, as is already done in parametrized NSBI.

- *Nuisance-parameter coverage.* A nominal reference must cover the support of all relevant systematic variations. Very large shape changes may require a broader mixture or a conditional reference flow.

The reference distribution can itself be optimized. Equal process normalization is a robust first choice, but the objective could incorporate worst-case effective sample size over components and parameter points. Mixtures of flows, heavier-tailed bases, or conditional proposals may be preferable in high dimensions. These choices affect computational efficiency, not the algebraic validity of Eq. (12), provided support is maintained and ratios are trained against the proposal actually used.

VIII. CONCLUSION

We have introduced hybrid neural simulation-based inference, a construction that combines a tractable normalizing-flow reference with high-precision classifier estimates of process-to-reference density ratios. Sampling the denominator class from the learned flow is essential: it makes the ratio the correction to that specific tractable proposal, so local flow error cancels in the ideal classifier limit. The resulting process densities can be evaluated, integrated, and sampled while inference retains the ratio structure of parametrized NSBI.

The hybrid representation supplies two statistical objects that are cumbersome in a ratio-only analysis: a deterministic weighted unbinned Asimov measure and an efficient process sampler for pseudo-experiments. The extended part of the Asimov likelihood is enforced by matching the total event weight to the expected yield, and ratio normalization on the same reference quadrature guarantees the fitted generating value for the one-parameter demonstration. The Asimov value of q_0 then predicts the estimator width and the noncentral toy distribution in the usual Wald regime.

The five-dimensional conference exercise is designed as an inference-level proof of concept, with analytic truth available for closure. Once its long training run is complete, the final manuscript will report the reference and hybrid-density diagnostics, the comparison of hybrid and analytic likelihood scans, the exact empirical Asimov closure, and the agreement between Asimov predictions and pseudo-experiment distributions. These tests will determine whether the proposed division of labor—generative coverage from a flow and precision corrections from ratios—delivers the expected practical advantage.

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